EDUX / ACADEMIC INTELLIGENCE

FULL EXAM ANALYSIS REPORT

QA Stress Fixture — variable subjects, scores and question counts

Synthetic validation paper · Mixed format · All Subjects · Generated 02 Nov 2026

STUDENT	ROLL NUMBER	EXAMINATION DATE	TOTAL MARKS
QA Fixture	QA-0001	2026-11-02	360

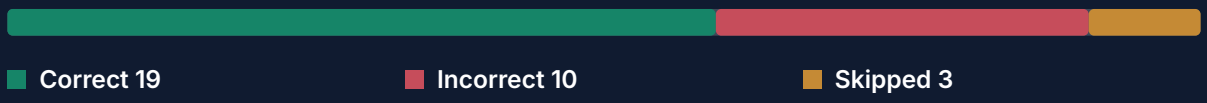
TOTAL MARKS OBTAINED

217

/ 360 marks

19 correct | 10 incorrect | 3 skipped





Correct 19 · Incorrect 10 · Skipped 3

SCORE	ACCURACY	RESPONSE DISTRIBUTION
60.3%	66.7%	19 correct | 10 incorrect | 3 skipped

PUBLISHED PERFORMANCE BRIEF

This is a synthetic edge-case fixture used only to validate pagination, mathematical typesetting, long labels, missing optional values and dynamic subject combinations. It is not a student report.

FAR / QA-EDGE-2026-01 · QA-EDGE-2026-01 · Source fingerprint 7CAE16BA · 2026-11-02T09:00:00.000Z

01 PERFORMANCE INTELLIGENCE

Performance intelligence

Response composition

32 questions in the paper



- Correct 19
- Incorrect 10
- Skipped 3

Mistake distribution

Count by published classification

Concept Error	5
Calculation Error	4
Formula Error	3
Time Pressure	2

Seven source classifications; rows are measured independently.

Difficulty analysis

Questions and published accuracy

Easy	10 Q · 82%
Medium	12 Q · 66%
Hard	10 Q · 41%

Accuracy values are reproduced from the payload.

Subject-wise performance

Accuracy by subject

Physics	67%
Chemistry	53%

Physics 121 / 180 Chemistry 96 / 180

Important performance signals

GRADE	RANK	PERCENTILE	BATCH RANK	COHORT SIZE	OVERALL SPEED	CONFIDENCE INDEX	READINESS SCORE
B	91	88.2	4	60	2.8	74	70

HEALTH SCORE	NEGATIVE MARKS	GUESS ATTEMPTS	ATTEMPT RATIO	TOTAL QUESTIONS	SUCCESS RATE	BADGE
79	9	2	90.7%	32	66.7%	Synthetic QA

02 MASTERY MAP

Subject and chapter mastery

Subject marks, accuracy and classification

SUBJECT	MARKS	ACCURACY	RANK	LEVEL	STRONG AREAS	WEAK AREAS
Physics	121 / 180	67%	2	Medium	Kinematics	Rotational Motion
Chemistry	96 / 180	53%	4	Hard	Mole Concept	Chemical Kinetics

Strong, weak and critical chapters by subject

SUBJECT	STRONG CHAPTERS	WEAK CHAPTERS	CRITICAL CHAPTERS
Physics	—	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Rotational Motion 5	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations · Rotational Motion 5
Chemistry	Chemical Kinetics 4 · Chemical Kinetics 8	Chemical Kinetics 2 · Chemical Kinetics 6	—

Chapter mastery · grouped by subject

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	MASTERY
Physics					
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	42%	4	3m	Critical
Chemistry					
Chemical Kinetics 2	Chemistry	48%	5	4m	Weak
Physics					
Rotational Motion 3	Physics	54%	6	5m	Average
Chemistry					
Chemical Kinetics 4	Chemistry	60%	7	6m	Strong
Physics					
Rotational Motion 5	Physics	66%	8	7m	Critical
Chemistry					
Chemical Kinetics 6	Chemistry	72%	9	8m	Weak
Physics					
Rotational Motion 7	Physics	78%	10	9m	Average

CHAPTER	SUBJECT	ACCURACY	MARKS	TIME	MASTERY
Chemistry					
Chemical Kinetics 8	Chemistry	84%	11	10m	Strong

03 PACE INTELLIGENCE

Time intelligence

Chapter time consumption · measured rows

CHAPTER	SUBJECT	SHARE OF TOTAL TIME	TIME CONSUMED	SHARE
Physics				
Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Physics	6%	3m	6%
Chemistry				
Chemical Kinetics 2	Chemistry	8%	4m	8%
Physics				
Rotational Motion 3	Physics	10%	5m	10%
Chemistry				
Chemical Kinetics 4	Chemistry	12%	6m	12%
Physics				
Rotational Motion 5	Physics	13%	7m	13%
Chemistry				
Chemical Kinetics 6	Chemistry	15%	8m	15%
Physics				
Rotational Motion 7	Physics	17%	9m	17%
Chemistry				
Chemical Kinetics 8	Chemistry	19%	10m	19%

Time-management metrics

AVERAGE / QUESTION	NAVIGATION COUNT	MANAGEMENT SCORE	FASTEST QUESTION
3	11	68	Q1 — Kinematics

FASTEST TIME	SLOWEST QUESTION	SLOWEST TIME
0.7	Q31 — Definite Integration	7.9

Sectional allocation and efficiency

SECTION	ALLOC.	USED	EFF.
Physics	60	48	125
Chemistry	60	42	143

04 COMPARISON & TRAJECTORY

Comparison and projection

Comparative performance

Benchmark scores

Previous test 198

Batch average 176

Current score 217 / 360

VS PREVIOUS TEST
+19

Comparison register

BENCHMARK	SCORE	PCT.	GRADE	REFERENCE
Previous test	198	84.1	B	—
Batch average	176	80.2	—	—

Projection and readiness

Score trajectory

QA 1 190

QA 2 217

Expected improvement +3.1% · Risk Medium

JEE PERCENTILE	NEET SCORE	EXPECTED AIR
91.1	—	21400

TARGET PROBABILITY	RISK LEVEL	EXPECTED IMPROVEMENT
77	Medium	+3.1%

Trajectory values

EXAMINATION / PROJECTION	SCORE
QA 1	190
QA 2	217

READINESS SCORE	CONFIDENCE INDEX	HEALTH SCORE	OVERALL ACCURACY	SUCCESS RATE	BADGE
70	74	79	66.7%	66.7%	Synthetic QA

05 QUESTION INTELLIGENCE

Question-by-question analysis

32 question-review rows supplied. Missing optional values are shown as —; full source explanations are never shortened.

REVIEW LEGEND

Correct · Incorrect · Skipped Each question presents its identity, classifications, answer comparison, marks, time, mistake category and complete solution.

Q1

Q1 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 1: $x^{1} + \sqrt{1} + \sum_{k=1}^{1} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	4	—

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_{0}^{1} \frac{x^{2}}{\sqrt{1-x^{2}}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q2

Q2 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 2: $x^2 + \sqrt{2} + \sum_{k=1}^{2} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	1m 54s

Mistake category · Concept Error

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q3 Q3 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 3: $x^3 + \sqrt{3} + \sum_{k=1}^{3} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	2m 36s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q4 Q4 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 4: $x^4 + \sqrt{4} + \sum_{k=1}^{4} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	3m 18s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

Q4 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

Q5

Q5 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 5: $x^5 + \sqrt{5} + \sum_{k=1}^{5} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	-1	4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q6 Q6 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 6: $x^{6} + \sqrt{6} + \sum_{k=1}^{6} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	4m 42s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^{2} = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^{2} + \beta^{2} \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q7 Q7 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion

DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 7: $x^{7} + \sqrt{7} + \sum_{k=1}^{7} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	5m 24s

EXPLANATION

A multi-line derivation:

Q7 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q8 Q8 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 8: $x^8 + \sqrt{8} + \sum_{k=1}^{8} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	-1	—

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Q9

Q9 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 9: $x^9 + \sqrt{9} + \sum_{k=1}^{9} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	6m 48s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q10

Q10 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion

DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 10: $x^{10} + \sqrt{10} + \sum_{k=1}^{10} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	1m 12s

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is

Q10 · solution continued

preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q11

Q11 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 11: $x^{11} + \sqrt{11} + \sum_{k=1}^{11} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	-1	1m 54s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q12 Q12 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 12: $x^{12} + \sqrt{12} + \sum_{k=1}^{12} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	2m 36s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

Q13 Q13 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 13: $x^{13} + \sqrt{13} + \sum_{k=1}^{13} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	3m 18s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

Q13 · solution continued

The substitution $x = \sin\theta$ gives $dx = \cos\theta\, d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q14 Q14 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 14: $x^{14} + \sqrt{14} + \sum_{k=1}^{14} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q15 Q15 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 15: $x^{15} + \sqrt{15} + \sum_{k=1}^{15} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	4	—

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q16 Q16 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 16: $x^{16} + \sqrt{16} + \sum_{k=1}^{16} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	5m 24s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is

Q16 · solution continued

also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

Q17

Q17 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 17: $x^{17} + \sqrt{17} + \sum_{k=1}^{17} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	-1	6m 06s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q18 Q18 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 18: $x^{18} + \sqrt{18} + \sum_{k=1}^{18} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	6m 48s

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q19 Q19 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 19: $x^{19} + \sqrt{19} + \sum_{k=1}^{19} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	1m 12s

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q19 · solution continued

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q20 Q20 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 20: $x^{20} + \sqrt{20} + \sum_{k=1}^{20} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	-1	1m 54s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Q21

Q21 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 21: $x^{21} + \sqrt{21} + \sum_{k=1}^{21} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	2m 36s

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q22 Q22 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	TYPE	STATUS
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 22: $x^{22} + \sqrt{22} + \sum_{k=1}^{22} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	4	—

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q23 Q23 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 23: $x^{23} + \sqrt{23} + \sum_{k=1}^{23} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	-1	4m

Mistake category · Concept Error

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and

Q23 · solution continued

$\lim_{n\to\infty}(1+1/n)^n = e$ ·

Q24 Q24 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 24: $x^{24} + \sqrt{24} + \sum_{k=1}^{24} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	4m 42s

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\int_0^1 \frac{x}{\sqrt{1-x^2}
```

. Neither fragment is guessed or silently repaired.

Q25 Q25 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 25: $x^{25} + \sqrt{25} + \sum_{k=1}^{25} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
A	B	4	5m 24s

EXPLANATION

The integral reduces directly:

Q25 · solution continued

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q26 Q26 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 26: $x^{26} + \sqrt{26} + \sum_{k=1}^{26} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	-1	6m 06s

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q27 Q27 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 27: $x^{27} + \sqrt{27} + \sum_{k=1}^{27} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	6m 48s

SOLUTION / EXPLANATION

EXPLANATION

A multi-line derivation:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}
```

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^{\infty} e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

Q28 Q28 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Chemical Kinetics	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 28: $x^{28} + \sqrt{28} + \sum_{k=1}^{28} k$. Options: A · 2 s B · √3 s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	4	1m 12s

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM

```
\frac{1}{2
```

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

Q28 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

MISTAKE NOTE

Used u = x^2 instead of u = x^2+1

Q29 Q29 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

DIFFICULTY	TYPE	STATUS
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 29: $x^{29} + \sqrt{29} + \sum_{k=1}^{29} k$. Options: —

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
—	—	-1	—

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

The integral reduces directly:

$$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$

The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

Q30 Q30 · Skipped

Skipped

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Hard	Assertion Reason	Skipped

QUESTION / OPTIONS

Question: Evaluate the expression for case 30: $x^{30} + \sqrt{30} + \sum_{k=1}^{30} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
B	C	4	2m 36s

SOLUTION / EXPLANATION

EXPLANATION

Using

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$

and noting $\alpha^2 + \beta^2 \le \sqrt{\gamma}$, the solution follows. Charge balance for $\mathrm{H_2SO_4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.

Q31 Q31 · Correct

Correct

SUBJECT	CHAPTER	TOPIC
Chemistry	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Easy	MCQ	Correct

QUESTION / OPTIONS

Question: Evaluate the expression for case 31: $x^{31} + \sqrt{31} + \sum_{k=1}^{31} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
C	D	4	3m 18s

EXPLANATION

A multi-line derivation:

Q31 · solution continued

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}

with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty}(1+1/n)^n = e$.

SOLUTION / HINT

Check the sign convention and the limits of integration carefully.

Q32 Q32 · Incorrect

Incorrect

SUBJECT	CHAPTER	TOPIC
Physics	Rotational Motion	Projectile motion
DIFFICULTY	**TYPE**	**STATUS**
Medium	Numerical	Incorrect

QUESTION / OPTIONS

Question: Evaluate the expression for case 32: $x^{32} + \sqrt{32} + \sum_{k=1}^{32} k$. Options: A · 2 s B · $\sqrt{3}$ s C · 4 s D · A very long alternative answer that must wrap without clipping the card

STUDENT ANSWER	CORRECT ANSWER	MARKS	TIME
D	A	-1	4m

Mistake category · Concept Error

SOLUTION / EXPLANATION

EXPLANATION

FLAGGED SOURCE — do not correct:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\frac{1}{2

is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced:

EXPRESSION FLAGGED FOR VERIFICATION — SOURCE SHOWN VERBATIM
\int_0^1 \frac{x}{\sqrt{1-x^2}

. Neither fragment is guessed or silently repaired.

06 LEARNING INTELLIGENCE

Learning intelligence

Published revision priorities

Priority revision register

PRIORITY TOPIC	CLASSIFICATION	TIMEFRAME
Definite Integration — substitution	Critical	48 hours
Chemical Kinetics — rate laws	High	5 days

Topic-level accuracy

TOPIC	SUBJECT	ACC.	LEVEL
Definite Integration	Physics	44%	Critical
Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Physics	51%	Weak
Bayes theorem	Physics	62%	Average
Coordination compounds	Chemistry	48%	Weak
Chemical Kinetics	Chemistry	82%	Strong
Mole Concept	Chemistry	91%	Strong

Mistake evidence

Q	SOURCE DETAIL
Q4	Misapplied substitution — used $u = x^2$ instead of $u = x^2+1$
Q11	Decimal error in $1/r^2$ computation

Weak chapters

NO.	WEAK CHAPTER
1	Calculus (44%)
2	Chemical Kinetics (48%)

Subject-wise revision sections

Priority topics grouped by subject

SUBJECT	PRIORITY TOPIC	CHAPTER	ACCURACY	CLASSIFICATION
Physics				
Physics	Definite Integration	—	44%	Critical
Physics	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations	51%	Weak
Physics	Bayes theorem	—	62%	Review
Chemistry				
Chemistry	Coordination compounds	—	48%	Weak

Recommended next actions

Published action register

ACTION	RESOURCE	DETAIL
PYQ	JEE Main 2024 — Integration (5 Qs)	PYQ Bank
Practice	20-question drill — substitution	20 · Medium
Mock test	Sectional mock — Chemistry	Sectional
Lecture	Kinetics masterclass	Video

Weak topics / published list

NO.	WEAK TOPIC
1	Definite Integration
2	Coordination compounds
3	Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions

PUBLISHED PERFORMANCE BRIEF

This is a synthetic edge-case fixture used only to validate pagination, mathematical typesetting, long labels, missing optional values and dynamic subject combinations. It is not a student report.

08 DATA INTEGRITY

Source data register

Source data register · 225 canonical payload values

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
meta.examId QA-EDGE-2026-01	meta.examName QA Stress Fixture — variable subjects, scores and question counts	meta.course Synthetic validation paper	meta.pattern Mixed format
meta.student QA Fixture	meta.rollNo QA-0001	meta.date 2026-11-02	meta.totalMarks 360
meta.duration 3 hrs	meta.sections[0] Physics	meta.sections[1] Chemistry	meta.examStatus Synthetic
hero.score 217	hero.percentage 60.3	hero.grade B	hero.rank 91
hero.percentile 88.2	hero.batchRank 4	hero.cohortSize 60	hero.overallAccuracy 66.7
hero.overallSpeed 2.8	hero.confidenceIndex 74	hero.readinessScore 70	hero.healthScore 79
hero.badge Synthetic QA	questionIntelligence.total 32	questionIntelligence.attempted 29	questionIntelligence.correct 19
questionIntelligence.incorrect 10	questionIntelligence.skipped 3	questionIntelligence.negativeMarks 9	questionIntelligence.guessAttempts 2
questionIntelligence.attemptRatio 90.7	attempt.elapsedSeconds 5400	subjects[0].score 121	subjects[0].maxMarks 180
subjects[0].accuracy 67	subjects[0].time 48	subjects[0].difficulty Medium	subjects[0].attempted 18

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
subjects[0].correct 12	subjects[0].weakAreas[0] Rotational Motion	subjects[0].strongAreas[0] Kinematics	subjects[1].score 96
subjects[1].accuracy 53	subjects[1].time 42	subjects[1].difficulty Hard	subjects[1].attempted 11
subjects[1].correct 7	subjects[1].weakAreas[0] Chemical Kinetics	subjects[1].strongAreas[0] Mole Concept	chapters[0].mastery Critical
chapters[1].chapter Chemical Kinetics 2	chapters[1].marks 5	chapters[1].attempted 73	chapters[1].mastery Weak
chapters[2].chapter Rotational Motion 3	chapters[2].accuracy 54	chapters[2].marks 6	chapters[2].attempted 76
chapters[2].mastery Average	chapters[3].chapter Chemical Kinetics 4	chapters[3].mastery Strong	chapters[4].chapter Rotational Motion 5
chapters[4].accuracy 66	chapters[4].marks 8	chapters[4].attempted 82	chapters[5].chapter Chemical Kinetics 6
chapters[5].accuracy 72	chapters[5].attempted 85	chapters[6].chapter Rotational Motion 7	chapters[6].accuracy 78
chapters[6].attempted 88	chapters[7].chapter Chemical Kinetics 8	chapters[7].accuracy 84	topics[0].topic Definite Integration
topics[0].mastery 44	topics[1].mastery 51	topics[2].topic Bayes theorem	topics[2].mastery 62
topics[3].topic Coordination compounds	mistakes[0].category Concept Error	mistakes[1].category Calculation Error	mistakes[2].category Formula Error
mistakes[3].category Time Pressure	mistakeList[0].q Q4	mistakeList[0].detail Misapplied substitution — used u = x^2 instead of u = x^2+1	mistakeList[1].q Q11
mistakeList[1].detail Decimal error in 1/r^2 computation	difficulty[0].level Easy	difficulty[0].time 2.1	difficulty[1].time 3.4
difficulty[2].accuracy 41	difficulty[2].time 5.2	timeIntelligence.fastestQuestion.q Q1 — Kinematics	timeIntelligence.fastestQuestion.time 0.7
timeIntelligence.slowestQuestion.q Q31 — Definite Integration	timeIntelligence.slowestQuestion.time 7.9	timeIntelligence.timeManagementScore 68	timeIntelligence.distribution[0].efficiency 125
timeIntelligence.distribution[1].efficiency 143	comparison.previousTest.label Previous test	comparison.previousTest.score 198	comparison.previousTest.percentile 84.1
comparison.batchAverage.label Batch average	comparison.batchAverage.score 176	comparison.batchAverage.percentile 80.2	comparison.deltas[0].label vs previous test
comparison.deltas[0].value +19	comparison.deltas[0].up true	recommendations.weakChapters[0] Calculus (44%)	recommendations.weakChapters[1] Chemical Kinetics (48%)
recommendations.priorityRevision[0].topic Definite Integration — substitution	recommendations.priorityRevision[0].timeframe 48 hours	recommendations.priorityRevision[1].topic Chemical Kinetics — rate laws	recommendations.priorityRevision[1].priority High
recommendations.priorityRevision[1].timeframe 5 days	recommendations.suggestedPYQs[0].title JEE Main 2024 — Integration (5 Qs)	recommendations.suggestedPYQs[0].source PYQ Bank	recommendations.practiceQuestions[0].title 20-question drill — substitution
recommendations.practiceQuestions[0].count 20	recommendations.mockTests[0].title Sectional mock — Chemistry	recommendations.mockTests[0].mode Sectional	recommendations.lectures[0].title Kinetics masterclass

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
recommendations.lectures[0].type Video	prediction.jeePercentile 91.1	prediction.expectedAIR 21400	prediction.expectedImprovement +3.1%
prediction.targetProbability 77	prediction.trajectory[0].exam QA 1	prediction.trajectory[0].score 190	prediction.trajectory[1].exam QA 2
questionReview[0].q Q1	questionReview[0].type MCQ	questionReview[0].status Correct	questionReview[0].time 0
questionReview[0].text Evaluate the expression for case 1: $x^{1} + \sqrt{1} + \sum_{k=1}^{1} k$.	questionReview[0].hint Check the sign convention and the limits of integration carefully.	questionReview[0].mistakeHint Used u = x^2 instead of u = x^2+1	questionReview[1].q Q2
questionReview[1].topic Projectile motion	questionReview[1].type Numerical	questionReview[1].status Incorrect	questionReview[1].marks -1
questionReview[1].time 1.9	questionReview[1].selected 1	questionReview[1].text Evaluate the expression for case 2: $x^{2} + \sqrt{2} + \sum_{k=1}^{2} k$.	questionReview[1].options[0] 2 s
questionReview[1].options[1] √3 s	questionReview[1].options[2] 4 s	questionReview[1].options[3] A very long alternative answer that must wrap without clipping the card	questionReview[2].q Q3
questionReview[2].type Assertion Reason	questionReview[2].status Skipped	questionReview[2].time 2.6	questionReview[2].text Evaluate the expression for case 3: $x^{3} + \sqrt{3} + \sum_{k=1}^{3} k$.
questionReview[3].time 3.3	questionReview[3].text Evaluate the expression for case 4: $x^{4} + \sqrt{4} + \sum_{k=1}^{4} k$.	questionReview[4].q Q5	questionReview[4].text Evaluate the expression for case 5: $x^{5} + \sqrt{5} + \sum_{k=1}^{5} k$.
questionReview[5].q Q6	questionReview[5].time 4.7	questionReview[5].text Evaluate the expression for case 6: $x^{6} + \sqrt{6} + \sum_{k=1}^{6} k$.	questionReview[6].q Q7
questionReview[6].time 5.4	questionReview[6].text Evaluate the expression for case 7: $x^{7} + \sqrt{7} + \sum_{k=1}^{7} k$.	questionReview[7].q Q8	questionReview[7].text Evaluate the expression for case 8: $x^{8} + \sqrt{8} + \sum_{k=1}^{8} k$.
questionReview[8].q Q9	questionReview[8].time 6.8	questionReview[8].text Evaluate the expression for case 9: $x^{9} + \sqrt{9} + \sum_{k=1}^{9} k$.	questionReview[9].q Q10
questionReview[9].time 1.2	questionReview[9].text Evaluate the expression for case 10: $x^{10} + \sqrt{10} + \sum_{k=1}^{10} k$.	questionReview[10].text Evaluate the expression for case 11: $x^{11} + \sqrt{11} + \sum_{k=1}^{11} k$.	questionReview[11].q Q12
questionReview[11].text Evaluate the expression for case 12: $x^{12} + \sqrt{12} + \sum_{k=1}^{12} k$.	questionReview[12].q Q13	questionReview[12].text Evaluate the expression for case 13: $x^{13} + \sqrt{13} + \sum_{k=1}^{13} k$.	questionReview[13].q Q14
questionReview[13].text Evaluate the expression for case 14: $x^{14} + \sqrt{14} + \sum_{k=1}^{14} k$.	questionReview[14].q Q15	questionReview[14].text Evaluate the expression for case 15: $x^{15} + \sqrt{15} + \sum_{k=1}^{15} k$.	questionReview[15].q Q16

SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE	SOURCE / VALUE
questionReview[15].text Evaluate the expression for case 16: $x^{16} + \sqrt{16} + \sum_{k=1}^{16} k$.	questionReview[16].q Q17	questionReview[16].time 6.1	questionReview[16].text Evaluate the expression for case 17: $x^{17} + \sqrt{17} + \sum_{k=1}^{17} k$.
questionReview[17].q Q18	questionReview[17].text Evaluate the expression for case 18: $x^{18} + \sqrt{18} + \sum_{k=1}^{18} k$.	questionReview[18].q Q19	questionReview[18].text Evaluate the expression for case 19: $x^{19} + \sqrt{19} + \sum_{k=1}^{19} k$.
questionReview[19].q Q20	questionReview[19].text Evaluate the expression for case 20: $x^{20} + \sqrt{20} + \sum_{k=1}^{20} k$.	questionReview[20].q Q21	questionReview[20].text Evaluate the expression for case 21: $x^{21} + \sqrt{21} + \sum_{k=1}^{21} k$.
questionReview[21].q Q22	questionReview[21].text Evaluate the expression for case 22: $x^{22} + \sqrt{22} + \sum_{k=1}^{22} k$.	questionReview[22].q Q23	questionReview[22].text Evaluate the expression for case 23: $x^{23} + \sqrt{23} + \sum_{k=1}^{23} k$.
questionReview[23].q Q24	questionReview[23].text Evaluate the expression for case 24: $x^{24} + \sqrt{24} + \sum_{k=1}^{24} k$.	questionReview[24].q Q25	questionReview[24].text Evaluate the expression for case 25: $x^{25} + \sqrt{25} + \sum_{k=1}^{25} k$.
questionReview[25].q Q26	questionReview[25].text Evaluate the expression for case 26: $x^{26} + \sqrt{26} + \sum_{k=1}^{26} k$.	questionReview[26].q Q27	questionReview[26].text Evaluate the expression for case 27: $x^{27} + \sqrt{27} + \sum_{k=1}^{27} k$.
questionReview[27].q Q28	questionReview[27].text Evaluate the expression for case 28: $x^{28} + \sqrt{28} + \sum_{k=1}^{28} k$.	questionReview[28].q Q29	questionReview[28].text Evaluate the expression for case 29: $x^{29} + \sqrt{29} + \sum_{k=1}^{29} k$.
questionReview[29].q Q30	questionReview[29].text Evaluate the expression for case 30: $x^{30} + \sqrt{30} + \sum_{k=1}^{30} k$.	questionReview[30].q Q31	questionReview[30].text Evaluate the expression for case 31: $x^{31} + \sqrt{31} + \sum_{k=1}^{31} k$.
questionReview[31].q Q32	questionReview[31].text Evaluate the expression for case 32: $x^{32} + \sqrt{32} + \sum_{k=1}^{32} k$.	— —	— —

Extended source values · 7 verbatim entries

SOURCE / VALUE	SOURCE / VALUE
hero.aiSummary This is a synthetic edge-case fixture used only to validate pagination, mathematical typesetting, long labels, missing optional values and dynamic subject combinations. It is not a student report.	chapters[0].chapter Thermodynamics and Statistical Mechanics — Advanced Applications in Multi-Stage Reversible Processes with Non-Ideal Corrections and Extended Derivations
topics[1].topic Non-equilibrium reversible heat transfer with variable coefficients and extended boundary conditions	questionReview[0].explanation The integral reduces directly: $$\int_0^1 \frac{x^2}{\sqrt{1-x^2}}\,dx = \frac{\pi}{4}.$$ The substitution $x = \sin\theta$ gives $dx = \cos\theta\,d\theta$ and the limits remain 0 to $\pi/2$. This derivation is intentionally long enough to test multi-line mathematical flow and vector rendering without shortening any source expression.

SOURCE / VALUE	SOURCE / VALUE
questionReview[1].explanation Using $$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6},$$ and noting $\alpha^2 + \beta^2 \leq \sqrt{\gamma}$, the solution follows. Charge balance for $\ce{H2SO4}$ and 1.6×10^{-19} C is preserved. Greek symbols, scripts, roots and scientific notation all remain in the source explanation.	questionReview[2].explanation A multi-line derivation: $$\begin{aligned} a &= b + c + \frac{1}{2} \\ d &= e + f - \sqrt{\lambda} \\ \therefore\; a+d &= b+c+e+f \\end{aligned}$$ with $\Delta E = q + w$ and $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$. The solution continues with $\int_0^\infty e^{-x^2}\,dx = \sqrt{\pi}/2$ and $\lim_{n\to\infty} (1+1/n)^n = e$.
questionReview[3].explanation FLAGGED SOURCE — do not correct: $\frac{1}{2$ is malformed and must be shown verbatim for verification. The following display expression is also deliberately unbalanced: $$\int_0^1 \frac{x}{\sqrt{1-x^2}$$. Neither fragment is guessed or silently repaired.	— —

MediXO EduX · Examination Intelligence

End of report · FAR / QA-EDGE-2026-01

Missing values are shown as —.